Soknarith (Soki) Sem

(401) 952–6819 — seminaia401@gmail.com — Seminaia — soknarith-sem — Worcester, MA

Summary

Ph.D. candidate in materials science and engineering with a background in chemical engineering and applied mathematics. Experienced in atomistic modeling, thermodynamics, and transport phenomena, with hands-on experience in lab-scale and manufacturing operations. Proficient in scientific computing tools including Python, MATLAB/Simulink, ASPEN Plus Dynamics, and R for data analysis.

Work Experience

Research Assistant / Graduate Teaching Assistant — **May 2024 to Present**
Worcester Polytechnic Institute – Worcester, MA

- Quantified defect energetics and transport behavior in oxide perovskites using DFT and thermodynamic modeling.
- Analyzed structure–property–process relationships across temperature and chemical potential regimes.
- Assisted in teaching undergraduate and graduate materials engineering courses by grading assignments and leading homework help sessions.

Chemical Process Engineer — **May 2023 to May 2024**
Knowles Precision Devices – New Bedford, MA

- Led the installation of new impregnation systems for high-voltage capacitors.
- Read P&ID and process flow diagrams to ensure proper equipment setup and process flow.
- Applied Lean/Six-Sigma/5S principles and statistical process controls in order to operate within KPI targets.
- Authored SOPs, technical documentation, safety protocols for material handling and process scale-up.
- Mentored technicians and operators on process procedures, safety protocols, and quality standards.
- Performed root cause analysis on batch failures, including CAPA implementation and process improvement suggestions.
- Modeled heater behavior in order to quantify the length of time required to reach target temperatures and pressures.

R&D Technician — **Mar 2023 to May 2023**
Factorial Energy – Woburn, MA

- Fabricated Li-metal batteries in controlled glovebox and dry-room environments.
- Reduced material waste by 20% through inventory system redesign.

Downstream Purification Engineer — **Jul 2022 to Mar 2023**
Eurofins Scientific – Framingham, MA

- Performed lab-scale separation and purification of proteins using AKTA Avant chromatography systems achieving 95–99% purity.
- Applied DOE methods to optimize buffer conditions and flow rate for maximum yield and purity at lab-scale.
- Prepared buffers, reagents, and packed chromatography columns for lab-scale purification runs.
- Conducted lab-scale TFF membrane filtration, UF/DF, and AKTA chromatography resin life-time studies to validate CIP protocols.
- Characterized purified proteins using SDS-PAGE, HPLC/MS, GC/MS, UV-Vis, and related analytical techniques.

Education

Ph.D. Materials Science & Engineering — **Exp: 2029**
Worcester Polytechnic Institute – Worcester, MA

M.S. Materials Science & Engineering — **2026**
Worcester Polytechnic Institute – Worcester, MA

B.S. Chemical Engineering; B.S. Applied Mathematics — **2022**
University of Rhode Island – Kingston, RI

Minors: Chemistry, Physics

Relevant Coursework: Advanced Process Engineering, Polymer Chemistry, Polymer Engineering, Heat and Mass Transfer, Fluid Mechanics, Separation Processes, Advanced Thermodynamics, Ceramic Engineering

Technical Skills

Atomistic Simulation:	DFT (VASP & GPAW), Molecular Dynamics (LAMMPS)
Scientific Computing:	Python, MATLAB/Simulink, ASPEN Plus Dynamics, Linux, LaTeX, R, Unicorn
Laboratory / Analytical:	Glovebox, clean-room, HPLC, AKTA FPLC, TFF, UF/DF, SEM/XRD, DSC/TGA, UV-Vis, Osmometry, HPLC/MS, GC/MS, pH meter, SDS-PAGE, KF coulometric titration
Process / Manufacturing:	SPC, DOE, Lean/Six-Sigma, Process optimization, SOP development, Tech Transfer (MSAT), cGMP